Test Cases – Smart Complaint Management System

Based on the supplied Software Requirements Specification (SRS)

Note: Actual Result is not included in this planned test-case sheet; Status is set to Not Executed. Test cases are derived from the SRS requirements and acceptance criteria.

Test Case ID	Requirement ID	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Priority	Status
TC01	FR-001	Submit complaint with valid details	Citizen is logged in and on the home screen.	1. Open New Complaint. 2. Enter a valid title and description. 3. Select category and severity. 4. Optionally attach an image and enable GPS. 5. Submit the complaint.	Title: Pothole on MG Road; description: Road surface damaged; category: Roads; severity: High; image: 2 MB.	Complaint is saved; a unique SCMS-YYYYMMDD-NNNNN ticket ID is shown; confirmation includes complaint details and timeline link.	High	Not Executed
TC02	FR-001	Validate mandatory complaint fields	Citizen is logged in and the complaint form is open.	1. Leave one or more mandatory fields (title, description, category, severity) empty. 2. Click Submit. 3. Review validation feedback.	Title: empty; description: valid; category: none; severity: High.	Submission is blocked and the missing mandatory fields are highlighted with validation messages.	High	Not Executed
TC03	FR-001	Reject oversized complaint image	Citizen is logged in and the complaint form is open.	1. Choose an image larger than 5 MB. 2. Attempt to attach it. 3. Observe the system response.	Image: 12 MB.	The image is rejected and the system displays a message that the file exceeds the 5 MB limit.	High	Not Executed
TC04	FR-001	Enforce complaint title and description limits	Citizen is logged in and the complaint form is open.	1. Enter a title longer than 120 characters. 2. Enter a description longer than 2000 characters. 3. Attempt to submit.	Title: 121+ characters; description: 2001+ characters.	The system does not accept title or description values beyond the specified limits.	High	Not Executed
TC05	FR-002	Suggest categories with sufficient NLP confidence	NLP model is deployed and active; citizen is submitting a complaint.	1. Enter a description that can be categorized. 2. Wait for NLP suggestions. 3. Review the top three suggestions and confidence scores. 4. Accept a suggested category or override it.	Description: water leakage from main pipe on 5th cross.	Top-three category suggestions and confidence scores are shown; suggestions meet the 70% confidence criterion; citizen can accept or override the category.	High	Not Executed
TC06	FR-002	Handle low-confidence NLP suggestions	NLP model is active and the complaint form is open.	1. Enter complaint text for which all category suggestions have confidence below 70%. 2. Review the system response. 3. Manually select a category.	All suggested categories have confidence below 70%.	The system asks the citizen to choose a category manually and allows the submission flow to continue.	Medium	Not Executed
TC07	FR-003	Generate a unique ticket ID after submission	Citizen is logged in and has completed all mandatory fields.	1. Submit a valid complaint. 2. Check the confirmation screen and ticket ID. 3. Submit another valid complaint and compare IDs.	Two valid complaint submissions.	Each successful submission displays a unique ticket ID in the SCMS-YYYYMMDD-NNNNN format.	High	Not Executed
TC08	FR-004	Route complaint to the appropriate department and agent	Complaint is submitted; routing rules and active agents are configured.	1. Submit a complaint with a known category and location. 2. Allow the system to apply category, location, and workload rules. 3. Inspect the assigned department and agent.	Category: Water Supply; location: Zone 3; multiple active agents.	Complaint is routed to the appropriate department and agent according to category, location, and agent workload.	High	Not Executed
TC09	FR-004	Route complaint when no department agent is available	Complaint submission is available; no agent is active in the target department.	1. Submit a valid complaint for the target department. 2. Allow routing to run. 3. Inspect the assignment and notifications.	Target department has no active agent.	Complaint is assigned to the supervisor queue and an escalation alert is sent.	High	Not Executed
TC10	FR-005	Verify complaint lifecycle and timestamped history	Citizen and agent accounts are available; complaint system is active.	1. Submit a complaint. 2. Move it through Assigned, In Progress, Pending Info, Resolved, and Closed as applicable. 3. Open the complaint status history.	Valid complaint; each lifecycle transition performed.	The lifecycle follows New → Assigned → In Progress → Pending Info → Resolved → Closed, with every transition logged with a timestamp.	High	Not Executed
TC11	FR-006	Trigger SLA warning when deadline is approaching	A complaint has an SLA deadline configured from category and severity rules.	1. Create or select a complaint with a 24-hour SLA. 2. Advance or simulate time until fewer than 2 hours remain. 3. Check alerts.	SLA: 24 hours; elapsed time: 22 hours.	An SLA warning alert is triggered when the complaint is within 2 hours of its deadline.	High	Not Executed
TC12	FR-006	Trigger alert when SLA is breached	A complaint has an SLA deadline and remains	1. Allow the complaint SLA deadline to pass. 2. Check system alerts and relevant notifications.	Complaint past its SLA deadline.	The system triggers an alert when the SLA is breached.	High	Not Executed

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			unresolved past that deadline.					
TC13	FR-007	Send notifications for status changes	Citizen has a complaint and notification channels are configured; SMS is opt-in.	1. Change the complaint status. 2. Check in-app and email notifications; check SMS for an opted-in citizen. 3. Measure notification delivery time.	Citizen opted in to SMS; status changed.	Notifications are sent through in-app, email, and opted-in SMS channels on status changes, within the specified notification timing criteria.	Medium	Not Executed
TC14	FR-008	Save resolution rating and feedback	Complaint is resolved and the citizen can provide feedback within 7 days.	1. Open the resolved complaint feedback prompt. 2. Select a rating from 1 to 5 stars. 3. Enter an optional comment. 4. Submit feedback. 5. Check the saved rating and CSAT.	Rating: 4 stars; comment: Fixed well.	Rating and optional comment are saved and the CSAT score is updated.	Medium	Not Executed
TC15	FR-008	Automatically close complaint when it remains unrated	Complaint is resolved and no rating has been submitted within 7 days.	1. Resolve the complaint. 2. Do not submit feedback during the 7-day period. 3. Check the complaint status after the period.	Resolved complaint with no rating for 7 days.	The unrated complaint is automatically closed after the defined 7-day feedback window.	Medium	Not Executed
TC16	FR-009	Escalate a stalled complaint	Supervisor is logged in; a complaint can be escalated.	1. Open a stalled complaint. 2. Choose Escalate. 3. Select the receiving team and provide a reason if prompted. 4. Confirm escalation. 5. Check queue, SLA, and notifications.	Complaint in progress and stalled; reason: No response from field team.	Ticket moves to the priority queue, its SLA is reset with a new SLA, and relevant parties are notified.	High	Not Executed
TC17	FR-010	Enforce role-based access control	Citizen, Agent, Supervisor, and Admin accounts exist.	1. Log in with each role. 2. Attempt actions permitted for that role. 3. Attempt an action not permitted for that role through the relevant interface/API. 4. Check the result.	Four role accounts; authorized and unauthorized actions.	Permissions are enforced for each role on every API call; unauthorized access is denied.	High	Not Executed
TC18	FR-011	Verify dashboard metrics refresh and accuracy	Complaints with known statuses, resolution times, SLA outcomes, and CSAT ratings exist.	1. Open the dashboard. 2. Record displayed open/closed counts, average resolution time, SLA compliance, and CSAT trends. 3. Wait for the 60-second refresh. 4. Compare metrics against known records.	Dataset with known complaint statuses and scores.	Dashboard metrics reflect stored data and refresh every 60 seconds.	Medium	Not Executed
TC19	FR-012	Verify immutable audit log records complaint actions	A complaint exists and users with different roles can perform actions.	1. Perform a complaint action such as assignment or status change. 2. Open the audit log. 3. Check actor ID and timestamp. 4. Attempt to alter or remove the recorded audit entry using available permissions.	Complaint action performed by a known user.	The action is recorded with actor ID and timestamp; the audit log is immutable and the action history is retrievable.	High	Not Executed
TC20	NFR-001	Verify concurrent-user and response-time targets	A test environment and load-testing setup are available.	1. Simulate 500 concurrent users. 2. Measure API response times at p99. 3. Measure page-load times at p95. 4. Record results against the specified targets.	500 concurrent users; API p99 target under 500 ms; page-load p95 target under 2 seconds.	The system supports 500 concurrent users with API response times under 500 ms (p99) and page loads under 2 seconds (p95).	High	Not Executed

SRS coverage note: This 20-case selection covers FR-001 through FR-012 and NFR-001. The SRS also specifies NFR-002 (TLS 1.3, AES-256, OWASP Top 10 mitigation) and NFR-003 (99.9% uptime, RPO 1 hour, RTO 30 minutes); these are not separately tested in this approximately 20-case selection.